MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTOR ASSEMBLY, MARK 80 MOD 0

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Mark 80 Mod 0 Rocket Motor Assembly described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Rocket Motor Assembly Mark 80 Mod 0 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

SPECIFICATIONS

Naval Air Systems Command (Code Ident 30003)

AS 3554	Propellant, N-44 (U)

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

673AS100	Rocket Motor, Mk 80 Mod 0
673AS101	WORD Rocket Motor

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Rocket Motor Mk 80 shall conform to Drawing 673AS100.

3.2 Item description. The Rocket Motor Mk 80 Mod 0 specified herein is comprised of the drogue release assembly and the Wind Oriented Rocket Deployment (WORD) Motor Assembly.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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3.3 Preproduction.

3.3.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the requirements of table I.

TABLE I. Propellant N-44 (AS 3554).

Property	Temp (°F)	Zero time		7-Year Equivalent			
				0 % Rel. Humidity		22 % Rel. Humidity	
		Min	Max	Min	Max	Min	Max
S_m (maximum stress) psi	77	116	193	116	202	116	208
E_m (elongation at S_m) %	77	30	50	14	50	26	50
Y_m (Young's modulus) psi	77	589	1093	589	1737	589	1745
B_r (strand burn rate at 1000 psi) in/sec	77	0.55	0.67	0.60	0.71	0.60	0.71

3.3.2 Environmental.

3.3.2.1 Drogue release assembly. The drogue release assembly shall not sustain any damage, initiate, or fail to operate and shall conform to the performance characteristics at the specified temperature after sequential exposure to the environments and conditions specified in table II.

TABLE II. Environmental conditions for drogue release assembly.

Condition	Unit number and sequence of test							
	1	2	3	4	5	6	7	8[1]
In-process inspection	x	x	x	x	x	x	x	x
Final inspection	x	x	x	x	x	x	x	x
Radiographic inspection	x	x	x	x	x	x	x	x
Three-foot drop	a	a	a					
Radiographic inspection	x	x	x					
Shock		b		a	a			
Vibration								
−40°			b					
70°						a		
160°							a	
Radiographic inspection	x	x	x	x	x	x	x	x
Shear pin								
−40°	b		c		b		b	
160°		c		b		b		

[1] Sample unit 8 retained for Government procuring activity examination and evaluation.

3.3.2.1.1 Vibration. The drogue release assembly shall not sustain damage, initiate, or fail to operate and shall conform to the performance characteristics after being subjected to not less than 8 hours of vibration along each of three mutually perpendicular axes in accordance with MIL-STD-2102 at −40° ± 5°F or 160° ± 5°F, as specified.

3.3.2.1.2 Three-foot drop. The drogue release assembly shall not sustain damage or fail to operate and shall conform to the performance characteristics after being subjected to a 3-foot drop as specified in 4.1.2.

3.3.2.1.3 Shock. The drogue release assembly shall not sustain damage or fail to operate and shall conform to the performance characteristics after being subjected to shock in accordance with MIL-STD-2102.

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3.4 Preproduction/production.

3.4.1 Propellant. The propellant shall conform to the requirements of AS 3554 when tested in accordance with MIL-A-85097.

3.4.2 Ballistic performance. The ballistic performance shall conform to table III.

TABLE III. Ballistic performance requirements.

Characteristic	Value Min	Value Max	Temperature (± 5°F)
Total impulse (lb-s)	30	40	-40, 160
Action time (ms)	180	290	-40, 160
Thrust (lb)	-	240	160
Ignition delay (ms)	-	10	-40, 160
Actuation force (lb)	15	25	-40, 160

3.4.2.1 Rise time. The rise time shall be not greater than 30 milliseconds (ms) from 10 percent of maximum thrust to 75 percent of maximum thrust on the rising portion of the thrust-time curve.

3.4.2.2 Drogue release assembly performance. Upon application of 1,800 psig to either inlet port, the release shall actuate within 400 to 1,800 psig and shall in no way hinder deployment of the WORD rocket motor in any configuration or load arrangement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction.

4.1.1 Vibration. In place of the procedure specified in the general specification, vibration shall be performed on the drogure release assemblies specified in table II at the specified temperature in accordance with MIL-STD-2102.

4.1.2 Three-foot drop. Three-foot drop shall be performed on the drogue release assemblies specified in table II in accordance with the following:

a. Ambient conditions.

b. Units shall be dropped 3 feet onto a 2-inch thick steel plate (minimum Brinell hardness of 207) embedded in a minimum of 2 feet of concrete. No unit shall be dropped more than once.

c. Drop units to impact as shown on figure 1.

4.2 Preproduction/production.

4.2.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: -40° ± 5°F
b. Hot: 160° ± 5°F

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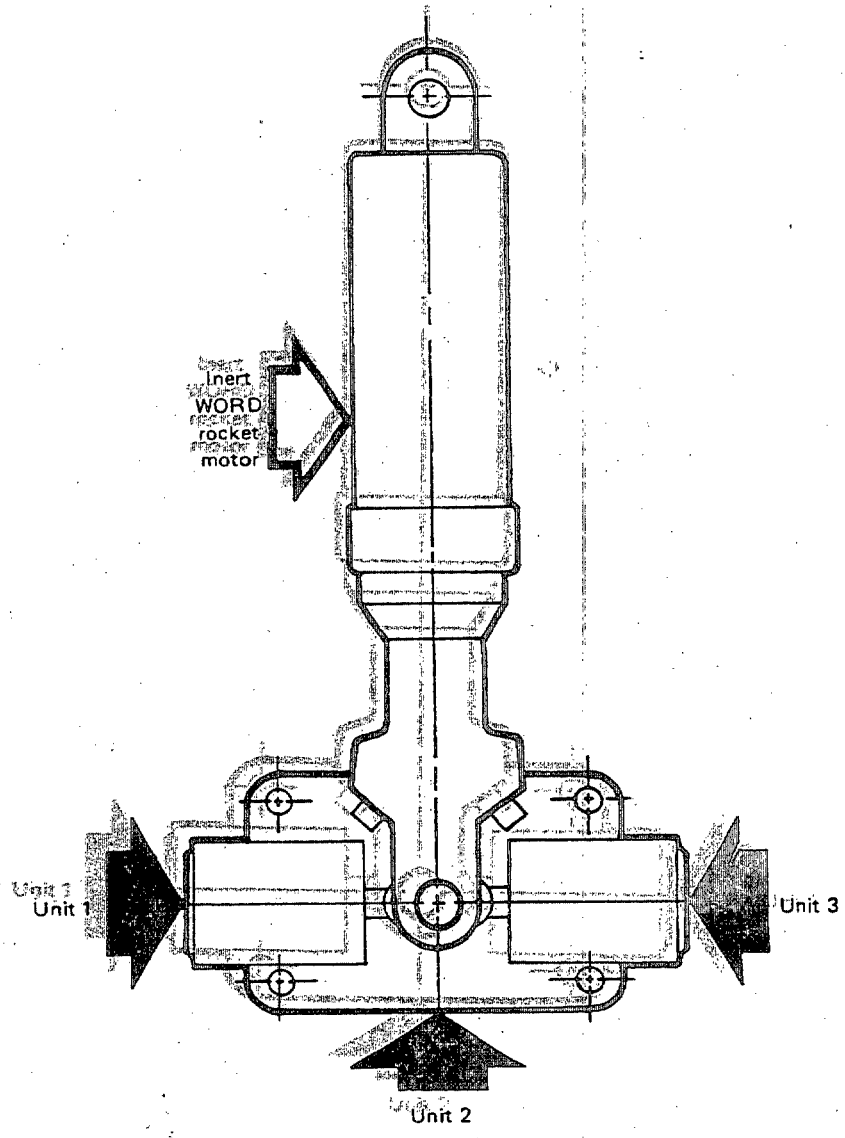


FIGURE 1. Impact points for 3-foot drop test.

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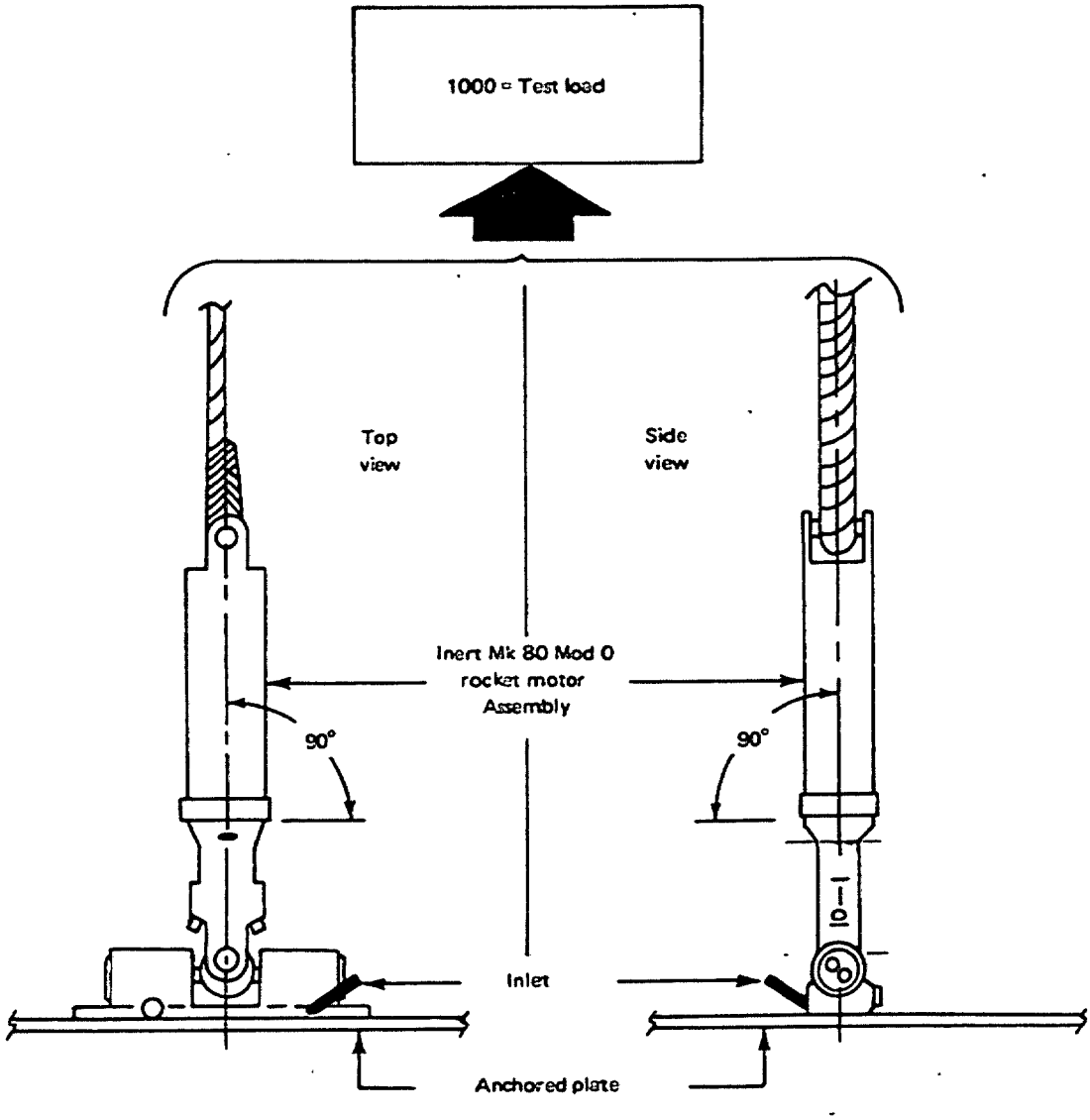


FIGURE 2. Test configuration.

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4.2.2 **Conditioning period.** Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: Not less than 3 hours
b. Temperature cycling: Not less than 4 hours

4.2.3 **Radiographic inspection.** All components or assemblies shall be radiographed in two views 90° apart.

4.2.4 **Drogue release assemblies.** Eight drogue release assemblies with CFE-fired WORD motors shall be supplied by the contractor for testing.

4.2.4.1 **Shear pin testing.** Shear pin testing shall be performed on the drogue release assemblies specified in table II in accordance with the following:

a. The test units shall be conditioned at the specified temperature in accordance with the general specification.

b. Test setup shall be as shown in figure 2.

c. Actuation pressure shall be regulated to 1,800 to 1,825 psig and pressurization rate shall be regulated to 30,000 to 50,000 psig/sec.

d. Independent shear pin testing shall be conducted on each inlet port.

5. PACKAGING

(See latest revision of MIL-A-85097.)

6. NOTES

6.1 **Intended use.** The Mk 80 Mod 0 Rocket Motor provides means for drogue parachute release and subsequent personnel parachute deployment in an aircrew automated escape system.

6.2 **Changes from previous issue.** Revision letters are not used to denote changes due to the extensiveness of the changes.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85097/4A(AS)	AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); MARK 8U MOD 0

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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